

Master Mathematics

Master thesis

ODE-Constrained Optimization for Quantitative MRI: A Single-Voxel Bloch Model

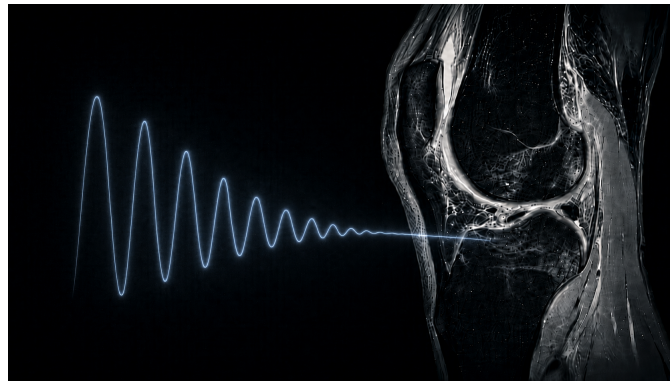
by

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July 31, 2026

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Abstract

This thesis studies an ODE-constrained inverse problem arising in quantitative magnetic resonance imaging. Tissue parameters are reconstructed from synthetic Bloch-equation data using a reduced formulation and a quadratic penalty formulation. The influence of the Forward Euler time discretization, the penalty parameter, and measurement noise is investigated through numerical experiments, while local identifiability is analyzed using contour plots and the Gauss–Newton Hessian. The results show that accurate reconstruction requires sufficiently small time steps, whereas increasing the penalty parameter mainly improves model consistency rather than parameter recovery. The Hessian analysis identifies weak sensitivity of the T_1 parameter as the dominant source of reconstruction error, and the noise experiments show that this deterministic error dominates over the tested noise levels.

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Summary

Magnetic Resonance Imaging (MRI) is widely used to produce images of the inside of the body. In quantitative MRI, the goal is not only to form an image, but also estimate physical tissue parameters from the measured signal. In this thesis, I study a simplified version of this problem for a single voxel. The parameters we want to estimate are the longitudinal relaxation time T_1 , the transverse relaxation time T_2 , and the proton density ρ . The dynamic of the magnetization in the voxel is described by the Bloch equation, an ordinary differential equation depending on these parameters. Since the parameters are not measured directly, they have to be recovered from the observed MRI signal. This makes the problem an inverse problem: instead of computing the signal from known parameters, we try to determine the parameters that could have produced the measured signal. Two numerical formulations are studied. In the reduced formulation, the Bloch equation is solved for each trial parameter vector, and the optimization is performed only over (T_1, T_2, ρ) . In the penalized formulation, both the magnetization trajectory and the parameters are treated as unknowns. The Bloch equation is then included through a penalty term, controlled by a parameter λ , which measures how strongly the computed trajectory is forced to follow the discrete model. A main question is how numerical choices affect the reconstruction. The Bloch equation is discretized using the Forward Euler method, so the time step h influences both accuracy and stability. The experiments show that, once h is within the stable time-step range, decreasing h improves the reduced reconstruction. However, the error does not disappear completely. Most of the remaining error comes from the T_1 component. The penalty experiments show that increasing λ reduces the residual in the discrete Bloch equation, meaning that the computed trajectory becomes more consistent with the model. However, this does not automatically improve the reconstructed parameters. In the tested examples, stronger model enforcement can increase the data misfit, and the parameter error is not monotone in λ . Thus, satisfying the model more accurately and reconstructing the parameters more accurately are related, but not equivalent. To explain this, the thesis studies the local geometry of the inverse problem using contour plots and the Gauss–Newton Hessian. These diagnostics show that the measured signal is much less sensitive to T_1 than to T_2 and ρ . This is because the signal directly observes the transverse magnetization, while T_1 mainly influences the longitudinal component and therefore enters the data more indirectly. The noise experiments confirm the same behaviour: measurement noise mostly increases the variability of the reconstructed T_1 .

Overall, the results show that the reconstruction error is not only a matter of choosing a small time step or a large penalty parameter. The local sensitivity structure of the inverse problem is crucial. For the chosen single-voxel experiment, T_1 is intrinsically harder to identify from the measured signal than T_2 and ρ .

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1 Introduction

Magnetic Resonance Imaging (MRI) is a non-invasive imaging technique widely used in medicine. It is based on the interaction between nuclear spins, primarily hydrogen protons, and externally applied magnetic fields [1]. When an object is placed in a strong static magnetic field B_0 , the spins tend to align with the field, resulting in a net magnetization vector. A radiofrequency (RF) pulse perturbs this magnetization from equilibrium. After excitation, the system relaxes back toward equilibrium through two main processes: longitudinal relaxation and transverse relaxation. These processes generate time-dependent signals, which are detected by receiver coils. The time evolution of the magnetization is described by the Bloch equations [2, 1]. The measured MRI signal is not a direct image, it is a superposition of contributions from the imaging volume. Magnetic field gradients are used to encode spatial information into the phase and frequency of the signal. The resulting data are sampled in k-space, the frequency-domain representation of the image, and standard MRI reconstruction methods use these data to form anatomical images, often through Fourier-based reconstruction techniques [1].

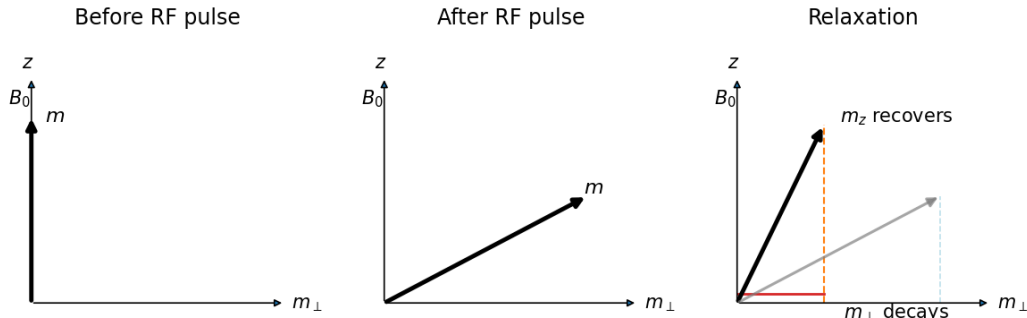


Figure 1.1: Schematic illustration of excitation and relaxation in MRI. Before the RF pulse, the net magnetization is aligned with the main magnetic field B_0 . The RF pulse tilts the magnetization away from the longitudinal axis, creating transverse magnetization. During relaxation, the longitudinal component m_z recovers while the transverse component $m_{\perp}$ decays.

Figure 1.1 illustrates the basic mechanism behind the two relaxation processes. Longitudinal relaxation describes the recovery of the m_z component, while transverse relaxation describes the decay of the transverse magnetization $m_{\perp}$. These two processes are characterized by different physical time constants, denoted by T_1 and T_2 . Beyond anatomical imaging, MRI can also be used to estimate quantitative tissue parameters. Quantitative

MRI aims to recover physical parameters such as the longitudinal relaxation time T_1 , the transverse relaxation time T_2 , and the proton density ρ from measured signals [3, 4, 5]. In this thesis, these three parameters are collected in the vector

$$\mathbf{p} = (T_1, T_2, \rho)^T.$$

The parameter T_1 describes how quickly the longitudinal magnetization recovers toward equilibrium, while T_2 characterizes the decay of magnetization in the transverse plane. The proton density ρ reflects the concentration of hydrogen nuclei and directly influences the signal amplitude. Together, these parameters determine the measured MRI signal and form the basis for quantitative tissue characterization.

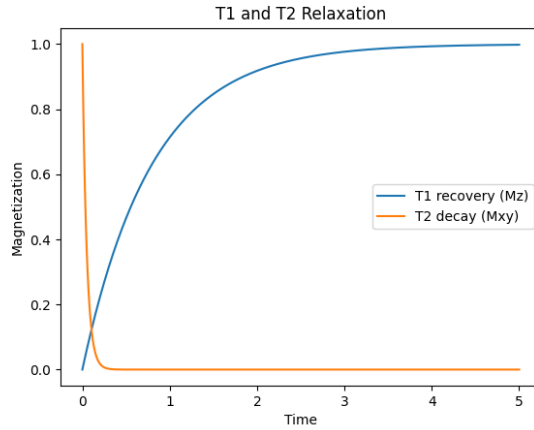


Figure 1.2: Typical longitudinal recovery and transverse decay in MRI. The longitudinal component recovers on a time scale governed by T_1 , while the transverse component decays on a time scale governed by T_2 .

Figure 1.2 complements the illustration in Figure 1.1 by showing the corresponding time behaviour. In many tissues, transverse relaxation occurs faster than longitudinal recovery, so typically $T_2 < T_1$. This difference in time scales is important for parameter reconstruction: the measured signal is directly related to the transverse magnetization, while information about T_1 enters more indirectly through the coupled Bloch dynamics. Recovering tissue parameters from measured MRI signals is therefore an inverse problem: the measured data are known, while the underlying parameters that produced them must be estimated. Such problems are often ill-conditioned or ill-posed, meaning that different parameter values may produce similar signals and small perturbations in the data may lead to large changes in the reconstruction [6, 7]. In quantitative MRI, these difficulties are amplified by the nonlinear dependence of the signal on the relaxation parameters, by measurement noise, by limited acquisition time, and by errors introduced through time discretization. The Bloch equations provide the forward model for this inverse problem. For a given parameter vector $\mathbf{p} = (T_1, T_2, \rho)^T$, they determine the time evolution of the magnetization and hence the predicted signal. Parameter reconstruction can therefore be formulated as an optimization problem constrained by the Bloch ODE [8, 9].

In this thesis, the term ODE-constrained optimization is used because the simplified single-voxel model depends only on time. More general MRI reconstruction problems, where the unknowns also vary over space, naturally lead to PDE-constrained or space-time constrained formulations. From the optimization point of view, the reconstruction can be formulated in two related ways. In the reduced formulation, the Bloch equation is solved for each trial parameter vector, so the optimization is carried out only over the unknown parameters. In an all-at-once formulation, both the magnetization trajectory and the parameters are treated as unknowns, and the Bloch equation is imposed as a model constraint. A quadratic penalty formulation relaxes this constraint by penalizing violations of the discrete Bloch equation [10, 11].

Here, the time step h controls the discretization of the Bloch dynamics, while the penalty parameter λ controls the strength of the model-consistency penalty. The main goal is to understand how these numerical choices interact with the sensitivity structure of the inverse problem. To achieve that, we address three related questions. The first question is studied by comparing reduced reconstructions for several Forward Euler time steps. The second question is studied by varying the penalty parameter and examining reconstruction errors, data misfits, ODE residuals, and L-curve behaviour. The final question is studied using local sensitivity analysis, the Gauss–Newton Hessian, contour plots of the reduced objective, and relative complex-valued noise experiments. The chapters are organized as follows. First, we introduce the single-voxel Bloch model, the measurement operator, and the reduced and penalized discrete inverse problems in Chapter 2. Then, Chapter 3 studies the interaction between time discretization and penalty enforcement. Chapter 4 analyzes local sensitivity and identifiability using contour plots and the Gauss–Newton Hessian. Finally, Chapter 5 investigates the effect of relative complex-valued measurement noise.

2 Mathematical formulation of forward and inverse problems

2.1 Forward Bloch problem

Let

$$\mathbf{m}(t) = (m_x(t), m_y(t), m_z(t))^T \in \mathbb{R}^3$$

denote the net magnetization of a single voxel. In MRI, the object is placed in a strong static magnetic field B_0 , which defines the longitudinal z -direction. The components m_x and m_y are called transverse components because they lie in the plane perpendicular to B_0 , while m_z is the longitudinal component because it is aligned with the z -axis. In the simplified rotating-frame model used in this thesis, the large static field B_0 determines the equilibrium direction but is not included explicitly in the model. Instead, we use an effective applied field

$$\mathbf{b} = (b_x, b_y, b_z)^T,$$

where b_x and b_y model transverse excitation and b_z represents an effective field offset in the longitudinal direction. In the numerical experiments, $\mathbf{b}$ is constant. If time-dependent RF pulses or gradients were included, the same notation would become $\mathbf{b}(t)$, and the Bloch matrix would become time-dependent. Recall from Chapter 1 that the unknown tissue parameter vector is ordered as

$$\mathbf{p} = (T_1, T_2, \rho)^T.$$

The time evolution of the magnetization is modeled by the Bloch equation, written here in affine form as

$$\frac{d\mathbf{m}}{dt} = \mathbf{A}(\mathbf{p})\mathbf{m}(t) + \mathbf{q}(\mathbf{p}), \quad (2.1)$$

with initial condition $\mathbf{m}(0) = \mathbf{m}_0$. Here, $\mathbf{A}(\mathbf{p})$ is defined as

$$\mathbf{A}(\mathbf{p}) = \begin{pmatrix} -1/T_2 & \gamma b_z & -\gamma b_y \\ -\gamma b_z & -1/T_2 & \gamma b_x \\ \gamma b_y & -\gamma b_x & -1/T_1 \end{pmatrix} \quad (2.2)$$

and the vector $\mathbf{q}(\mathbf{p}) = (0, 0, \rho/T_1)^T$ represents the relaxation towards equilibrium magnetization. We can interpret this as the following. The diagonal entries describe relaxation: the transverse components decay with rate $1/T_2$, while the longitudinal component relaxes with rate $1/T_1$. The off-diagonal terms proportional to $\gamma\mathbf{b}$ describe precession induced by the applied effective field. The source term ρ/T_1 in the third component

drives m_z toward the equilibrium value ρ . These relaxation times are tissue specific and carry quantitative information that is the target of quantitative MRI (qMRI). The parameter γ is denoting a gyromagnetic ratio, which for the remaining of the research we set to $\gamma = 1$, for convenience.

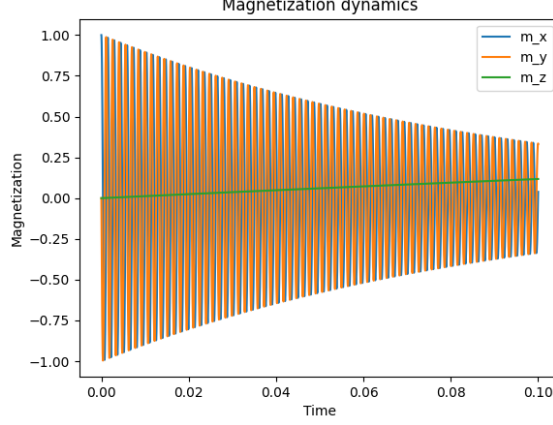


Figure 2.1: Time evolution of the magnetization components in the simplified single-voxel Bloch model.

Figure 2.1 shows the different time scales present in the Bloch dynamics. The transverse components m_x and m_y oscillate and decay due to transverse relaxation, while the longitudinal component m_z evolves toward its equilibrium value determined by the proton density ρ . This separation is important for the inverse problem: the measured signal depends directly on the transverse components, while information about T_1 enters more indirectly through the coupled evolution of the full magnetization vector. This feature plays an important role in the identifiability analysis in Chapter 4.

2.2 Inverse problem and measurement model

Following the signal model used in MR-STAT [5], the receiver signal can be written as a spatial integral of the transverse magnetization weighted by proton density and receiver sensitivity. In a Fourier-encoded acquisition this gives,

$$s(t) = \int_{\Omega} \rho(\mathbf{r}) (m_x(\mathbf{r}, t) + im_y(\mathbf{r}, t)) B_1^-(\mathbf{r}) e^{-2\pi i \mathbf{k}(t) \cdot \mathbf{r}} d\mathbf{r}.$$

where $B_1^-(\mathbf{r})$ denotes the receiver coil sensitivity. Also, $\mathbf{k}(t) \in \mathbb{R}^3$ denotes the k-space sampling trajectory generated by the gradient fields. K-space is the frequency-domain representation of the image: each measured data point corresponds to one sampled spatial frequency. The vector $\mathbf{k}(t)$ therefore does not store the data itself, but specifies which spatial frequency is being measured at time t . The factor $e^{-2\pi i \mathbf{k}(t) \cdot \mathbf{r}}$ describes the corresponding spatial phase modulation. [1, 5]. Here $\mathbf{r} = (x, y, z) \in \Omega \subset \mathbb{R}^3$

denotes spatial position and the notation $\mathbf{k}(t) \cdot \mathbf{r}$ denotes the Euclidean inner product in $\mathbb{R}^3$. This formulation reflects the fact that MRI measurements are spatially encoded and collected in the frequency domain. For the purpose of parameter estimation, we consider a simplified single-voxel and single-coil model. This means that we estimate one parameter vector $\mathbf{p}$ for one spatial location, rather than a parameter field over the whole image, and the receive sensitivity is ignored. Assuming unit receive sensitivity $B_1^- = 1$ and neglecting spatial encoding, the measurement model reduces to a direct observation of the transverse magnetization. In this case, the signal can be written as

$$d(t, \mathbf{p}) = \rho(m_x(t, \mathbf{p}) + im_y(t, \mathbf{p}))$$

where $\mathbf{m}(t, \mathbf{p}) = (m_x(t, \mathbf{p}), m_y(t, \mathbf{p}), m_z(t, \mathbf{p}))^T$ is the solution of the Bloch equations corresponding to the parameter vector $\mathbf{p}$. This expression can be written more compactly using a measurement operator. For a fixed parameter vector $\mathbf{p}$, we define the row vector

$$C := (1 \quad i \quad 0)$$

and the corresponding measurement operator

$$\mathcal{C}(\mathbf{m}, \mathbf{p}) := \rho C \mathbf{m} = \rho(m_x + im_y).$$

Thus, the single-voxel signal is

$$d(t, \mathbf{p}) = \mathcal{C}(\mathbf{m}(t, \mathbf{p}), \mathbf{p}).$$

This formulation makes explicit that the measurement process consists of extracting and scaling the transverse components of the magnetization. At the continuous level, we denote the observed signal by $d(t)$. In the noise-free synthetic setting, this signal is generated from a true parameter vector

$$\mathbf{p}^* = (T_1^*, T_2^*, \rho^*)^T.$$

More generally, one may write

$$d(t) = \mathcal{C}(\mathbf{m}(t, \mathbf{p}^*), \mathbf{p}^*) + \varepsilon(t),$$

where $\varepsilon(t) \in \mathbb{C}$ represents measurement noise. The goal of quantitative MRI is to estimate the parameter vector $\mathbf{p}$ from the measured signal. At the continuous level, this can be viewed as an ODE-constrained optimization problem, analogous to the PDE-constrained formulations commonly used in inverse problems [9, 8],

$$\min_{\mathbf{m}, \mathbf{p}} \frac{1}{2} \int_0^T |\mathcal{C}(\mathbf{m}(t), \mathbf{p}) - d(t)|^2 dt$$

with $|\cdot|$ being the complex modulus, subject to (2.1). In this formulation, both the trajectory $\mathbf{m}$ and the parameter vector $\mathbf{p}$ are unknowns, while the Bloch equation is

imposed as a constraint. Equivalently, one may solve the Bloch equation for each fixed parameter vector $\mathbf{p}$. This eliminates the state variable and gives the reduced objective

$$J_\infty(\mathbf{p}) = \frac{1}{2} \int_0^T |\mathcal{C}(\mathbf{m}(t, \mathbf{p}), \mathbf{p}) - d(t)|^2 dt.$$

This reduced formulation is the starting point for the standard reconstruction method used in Chapter 3. Figure 2.2 illustrates how each parameter affects the simplified single-voxel signal when the other two parameters are kept fixed. This is useful for understanding the inverse problem before introducing the numerical reconstruction. The parameter T_1 mainly affects the longitudinal recovery $m_z(t)$, whereas the measured signal depends directly on the transverse components $m_x(t)$ and $m_y(t)$. The parameter T_2 directly changes the decay rate of the transverse signal, and ρ rescales the signal amplitude. Therefore, the figure already suggests that T_2 and ρ have a more direct influence on the measured data than T_1 . This observation will be quantified later using the Hessian analysis in Chapter 4.

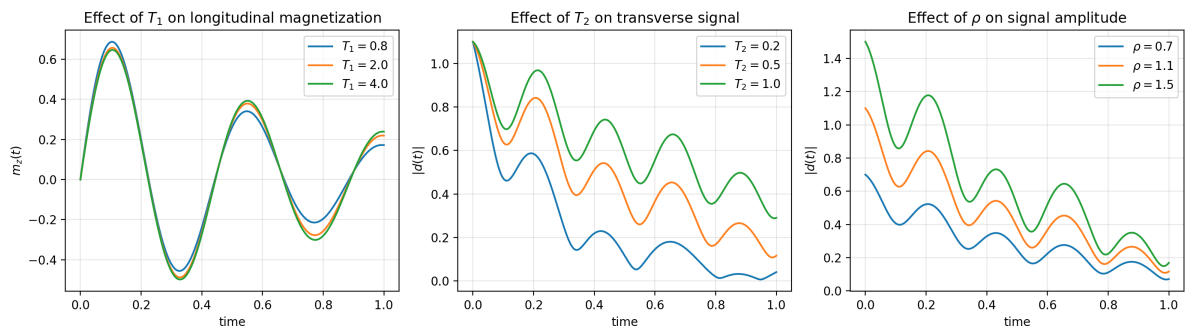


Figure 2.2: Effect of changing T_1 , T_2 , and ρ in the simplified single-voxel Bloch model. In each panel, one parameter is varied while the other two are fixed at $\mathbf{p}^* = (2.0, 0.5, 1.1)^T$.

2.3 Discrete formulation

In practice, the Bloch equation is solved numerically on a time grid. Let

$$0 = t_0 < t_1 < \dots < t_N = T$$

be the time grid used for the Forward Euler discretization, with $t_n = nh$ and $h = T/N$. This grid should be distinguished from the measurement times at which the data are sampled. In the numerical experiments of this thesis, the predicted signal is evaluated on the measurement grid used for the data; when the solver grid and the measurement grid do not coincide, the computed trajectory is evaluated at the measurement times by interpolation. We denote by $\mathbf{m}_n \approx \mathbf{m}(t_n, \mathbf{p})$ the Forward Euler approximation of the

magnetization. Forward Euler is a first-order explicit time-stepping method for ordinary differential equations [12]. It satisfies

$$\mathbf{m}_{n+1}(\mathbf{p}) = \mathbf{m}_n(\mathbf{p}) + h(\mathbf{A}(\mathbf{p}, t_n)\mathbf{m}_n(\mathbf{p}) + \mathbf{q}(\mathbf{p}, t_n)). \quad (2.3)$$

For an arbitrary discrete trajectory $\mathbf{M} = (\mathbf{m}_0, \dots, \mathbf{m}_N)$, we define the discrete ODE residual

$$\mathbf{R}_n(\mathbf{M}, \mathbf{p}) = \frac{\mathbf{m}_{n+1} - \mathbf{m}_n}{h} - (\mathbf{A}(\mathbf{p})\mathbf{m}_n + \mathbf{q}(\mathbf{p})), \quad n = 0, \dots, N-1. \quad (2.4)$$

If the discrete states exactly satisfy the Forward Euler discretization of the Bloch equation, then

$$\mathbf{R}_n(\mathbf{M}, \mathbf{p}) = \mathbf{0}, \quad n = 0, \dots, N-1.$$

The discrete measured data are denoted by $d_n = d(t_n)$, or, in the noisy case,

$$d_n = d(t_n) + \varepsilon_n.$$

Thus the data vector is $\mathbf{d} = (d_0, \dots, d_N)$. Once the problem has been discretized, there are several ways to solve the resulting finite-dimensional inverse problem. In the reduced approach, the state trajectory is not treated as an independent unknown. Instead, for every trial parameter vector $\mathbf{p}$, the Forward Euler recursion is solved forward in time, producing a discrete trajectory

$$\mathbf{M}_h(\mathbf{p}) = (\mathbf{m}_0(\mathbf{p}), \dots, \mathbf{m}_N(\mathbf{p})).$$

This is the standard reduced viewpoint in constrained inverse problems: the state equation is solved for each parameter value, resulting in an objective depending only on the parameters [9, 8]. The optimization is then carried out only over the parameter vector

$$\min_{\mathbf{p}} J_{h,\infty}(\mathbf{p}), \quad J_{h,\infty}(\mathbf{p}) = \frac{h}{2} \sum_{n=0}^N |\mathcal{C}(\mathbf{m}_n(\mathbf{p})) - d_n|^2.$$

The discrete sum is used as a Riemann approximation of the continuous time integral in J_∞ , hence the factor h also appears in front. In the discrete reduced formulation, every admissible parameter vector determines its own state trajectory through the Forward Euler method. Alternatively, one may keep both the state trajectory $\mathbf{M}$ and the parameter vector $\mathbf{p}$ as unknowns. This leads to an all-at-once constrained problem, where the residual equations $\mathbf{R}_n(\mathbf{M}, \mathbf{p}) = \mathbf{0}$ are imposed as constraints. Such reduced and all-at-once formulations are standard alternatives for optimization problems constrained by differential equations [9, 8]. The quadratic penalty method, introduced in the next subsection, is a relaxation of this all-at-once viewpoint: instead of enforcing $\mathbf{R}_n = \mathbf{0}$ exactly, violations of the discrete Bloch equation are penalized in the objective. This gives a practical way to balance data fidelity and model consistency.

Stability of Forward Euler. For any time discretization, stability is an important consideration. For Forward Euler, which is explicit, this leads to a restrictive time-step condition. The stability condition below is the classical absolute-stability condition for Forward Euler applied to the scalar test equation [12]. For the scalar test equation

$$\dot{y} = \mu y,$$

Forward Euler gives $y_{n+1} = (1 + h\mu)y_n$. For a linear system, this condition is applied to all eigenvalues μ_j of the system matrix. Thus, for a constant Bloch matrix $A(\mathbf{p})$, the time step should satisfy

$$|1 + h\mu_j| \leq 1$$

for all eigenvalues μ_j of $A(\mathbf{p})$. In the Bloch model considered here, the real parts of the eigenvalues are negative for all admissible parameters. This follows from the structure of $A(\mathbf{p})$, where the precession terms form a skew-symmetric part, while the relaxation terms contribute the symmetric part

$$\frac{1}{2} (A(\mathbf{p}) + A(\mathbf{p})^T) = \text{diag} \left(-\frac{1}{T_2}, -\frac{1}{T_2}, -\frac{1}{T_1} \right).$$

Since the admissible parameter set satisfies $T_1 > 0$ and $T_2 > 0$, this symmetric part is negative definite. Therefore the eigenvalues may be written as $\mu_j = a_j + ib_j$ with $a_j < 0$. The stability condition then gives

$$h \leq \frac{-2a_j}{a_j^2 + b_j^2}.$$

This gives the critical Forward Euler time step

$$h_{\text{crit}} = \min_j \frac{-2a_j}{a_j^2 + b_j^2}, \quad (2.5)$$

where the minimum is taken over all eigenvalues of the Bloch matrix. For $h < h_{\text{crit}}$, the Forward Euler discretization is expected to remain stable for the linearized constant-coefficient system. In the numerical experiments, this value is used only as a diagnostic threshold for the chosen parameter values and effective field.

2.4 Quadratic Penalty Formulation for the Bloch ODE

Quadratic penalty methods replace a hard equality constraint by a penalty term in the objective and recover the constrained solution in the large-penalty limit under suitable regularity assumptions [10, 11]. The penalty formulation treats both $\mathbf{M}$ and $\mathbf{p}$ as unknowns and penalizes violations of (2.4). Define the data misfit

$$D_h(\mathbf{M}, \mathbf{p}) = \frac{h}{2} \sum_{n=0}^N |\mathcal{C}(\mathbf{m}_n, \mathbf{p}) - d_n|^2, \quad (2.6)$$

and the ODE residual term

$$\Psi_h(\mathbf{M}, \mathbf{p}) = \frac{h}{2} \sum_{n=0}^{N-1} \|\mathbf{R}_n(\mathbf{M}, \mathbf{p})\|_2^2. \quad (2.7)$$

Here $|\cdot|$ denotes the complex modulus, since the measured signal is complex-valued. The notation $\|\cdot\|$ denotes the Euclidean norm in $\mathbb{R}^3$, since each ODE residual $\mathbf{R}_n(\mathbf{M}, \mathbf{p})$ is a real three-dimensional vector. The factors h in both terms appear because the sums approximate time integrals. The fully discrete quadratic penalty objective is

$$J_{h,\lambda}(\mathbf{M}, \mathbf{p}) = D_h(\mathbf{M}, \mathbf{p}) + \lambda \Psi_h(\mathbf{M}, \mathbf{p}), \quad \lambda > 0. \quad (2.8)$$

The parameter λ controls how strongly the discrete Bloch equation is enforced. Small λ allows the state trajectory to fit the data while violating the model equation. Large λ forces the trajectory to satisfy the discrete dynamics more accurately. In the limit $\lambda \rightarrow \infty$, and provided that the constrained problem has a regular local minimizer, bounded Lagrange multipliers, and no degeneracy of the active constraint, minimizers of the quadratic penalty problem (2.8) approach minimizers of the equality-constrained discrete problem [10]. The penalty formulation should be interpreted as a relaxation of the exact discrete model constraint. This can be useful when the discrete model is only an approximation of the underlying dynamics, for example because of time discretization error or simplified modelling assumptions. The parameter λ then controls the compromise between fitting the observed signal and enforcing consistency with the discrete Bloch equation. To minimize J , we compute the partial derivatives with respect to the state trajectory M and the parameters $\mathbf{p}$. The gradient with respect to $\mathbf{m}$ is given by

$$\nabla_{\mathbf{M}} J_{h,\lambda} = \nabla_{\mathbf{M}} D_h + \lambda h \sum_{n=0}^{N-1} (\partial_{\mathbf{M}} \mathbf{R}_n)^T \mathbf{R}_n.$$

Similarly, the gradient with respect to $\mathbf{p}$ is

$$\nabla_{\mathbf{p}} J_{h,\lambda} = \nabla_{\mathbf{p}} D_h + \lambda h \sum_{n=0}^{N-1} (\partial_{\mathbf{p}} \mathbf{R}_n)^T \mathbf{R}_n.$$

This formula can be evaluated efficiently once the residuals R_n and the states M_n are known. In the numerical experiments, we approximate this derivative using finite differences as a first step, and then optionally switch to the analytic expression for improved efficiency and accuracy. In ill-posed inverse problems, very high accuracy in the constraints is not always needed; it may be sufficient to enforce the components of the model that are relevant for the identifiable part of the parameter [11]. Whether an intermediate value of λ gives the best reconstruction is an empirical question investigated in Chapter 3.

3 Interaction of Discretization and Penalization

The formulation in Chapter 2 contains two approximations that are important in practice. The first is the time discretization of the Bloch equation, and the second is the replacement of the hard discrete ODE constraint by a quadratic penalty term. This chapter studies these two effects separately and then jointly. The aim is not to prove a general convergence theorem for all Bloch inverse problems, but to understand, in the present single-voxel setting, when the numerical reconstruction is dominated by discretization error, when it is dominated by penalty/model-consistency error, and when the remaining error is caused by the sensitivity structure of the inverse problem itself. Using the notation from Chapter 2, the reduced continuous objective is denoted by J_∞ , the time-discrete reduced objective by $J_{h,\infty}$, the continuous penalized objective by J_λ , and the fully discrete penalized objective by $J_{h,\lambda}$. These objects form the approximation hierarchy

$$J_\infty \longrightarrow J_{h,\infty} \longrightarrow J_{h,\lambda}.$$

when one first discretizes the Bloch equation and then relaxes the discrete ODE constraint. Alternatively, one may first introduce a continuous penalty and then discretize,

$$J_\infty \longrightarrow J_\lambda \longrightarrow J_{h,\lambda}.$$

Both paths lead to the same fully discrete penalized functional $J_{h,\lambda}$, but the intermediate approximations emphasize different sources of error. This motivates the question whether, for the tested values, refining the time step h and increasing the penalty parameter λ lead to consistent reconstructions.

3.1 Discretization of the Reduced Problem

We begin with the continuous reduced inverse problem. This reduced formulation is obtained by eliminating the state equation, a standard approach in inverse problems constrained by differential equations [9, 8]. For each parameter vector $\mathbf{p}$, let $\mathbf{m}(t, \mathbf{p})$ denote the solution of the Bloch equation with the prescribed initial condition. The ideal reduced objective is

$$J_\infty(\mathbf{p}) = \frac{1}{2} \int_0^T |\mathcal{C}(\mathbf{m}(t, \mathbf{p})) - d(t)|^2 dt,$$

where $\mathcal{C}(\mathbf{m})$ is the single-voxel measurement operator introduced in Chapter 2. In the synthetic experiments, $d(t)$ is generated from the reference parameter vector $\mathbf{p}^*$ using a

high-accuracy reference solver DOP853. Using the Forward Euler discretization (2.3), the corresponding reduced discrete objective is

$$J_{h,\infty}(\mathbf{p}) = \frac{h}{2} \sum_{n=0}^N |\mathcal{C}(\mathbf{m}_n(\mathbf{p}), \mathbf{p}) - d_n|^2.$$

The subscript ∞ indicates that no finite penalty relaxation is used: for each parameter vector $\mathbf{p}$, the state $\mathbf{m}(t, \mathbf{p})$ is determined by the Bloch equation rather than treated as an independent optimization variable. As discussed in Section 2.3, the Forward Euler approximation is reliable only when the time step lies in the stability interval. Then, the method is first-order accurate: for fixed $\mathbf{p}$,

$$\max_{0 \leq n \leq N} \|\mathbf{m}_n(\mathbf{p}) - \mathbf{m}(t_n, \mathbf{p})\| \leq Gh,$$

where G is a constant independent of h for sufficiently small h . This is the standard first-order global error estimate for Forward Euler, provided the solution is sufficiently smooth and the time step lies in the stable range [12]. For the following consistency statement, we assume that the admissible parameter set $\mathcal{P}$ is compact and that T_1 and T_2 are bounded away from zero. On this set, the maps $\mathbf{A}(\mathbf{p}, t)$ and $\mathbf{q}(\mathbf{p}, t)$ are assumed to be continuously differentiable in $\mathbf{p}$ and continuous in t . We also assume that the corresponding exact solution $\mathbf{m}(t, \mathbf{p})$ is sufficiently smooth in time on $[0, T]$, and that the measurement map $\mathcal{C}$ is evaluated on bounded trajectories. These assumptions are satisfied in the present single-voxel Bloch model as long as the parameters remain inside the prescribed bounds. The entries of $\mathbf{A}(\mathbf{p})$ and $\mathbf{q}(\mathbf{p})$ depend on the parameters only through $1/T_1$, $1/T_2$, and ρ/T_1 . Since the admissible set bounds T_1 and T_2 away from zero, these functions are smooth and bounded on the parameter set. Moreover, the effective field $\mathbf{b}$ is constant in the numerical experiments, so the Bloch equation is a linear ODE with bounded coefficients. Therefore the solution $\mathbf{m}(t, \mathbf{p})$ exists, is smooth in time, and depends smoothly on $\mathbf{p}$ on the finite time interval $[0, T]$. The measurement operator is linear in $\mathbf{m}$ and smooth in $\mathbf{p}$, so the reduced objective is well-defined and smooth on the admissible parameter set. We now justify formally that $J_{h,\infty}$ is a first-order approximation of J_∞ . Define

$$g(t, \mathbf{p}) = \frac{1}{2} |\mathcal{C}(\mathbf{m}(t, \mathbf{p}), \mathbf{p}) - d(t)|^2.$$

Then

$$J_\infty(\mathbf{p}) = \int_0^T g(t, \mathbf{p}) dt.$$

Since $g(\cdot, \mathbf{p})$ is continuously differentiable in time, the corresponding Riemann sum satisfies

$$h \sum_{n=0}^N g(t_n, \mathbf{p}) = J_\infty(\mathbf{p}) + O(h).$$

The discrete objective does not use the exact values $\mathbf{m}(t_n, \mathbf{p})$, but the Forward Euler values $\mathbf{m}_n(\mathbf{p})$. By the first-order convergence of Forward Euler in the stable time-step range,

$$\mathbf{m}_n(\mathbf{p}) = \mathbf{m}(t_n, \mathbf{p}) + O(h)$$

uniformly in n . Since $\mathcal{C}$ is linear in $\mathbf{m}$, this implies

$$\mathcal{C}(\mathbf{m}_n(\mathbf{p})) = \mathcal{C}(\mathbf{m}(t_n, \mathbf{p})) + O(h).$$

Therefore,

$$|\mathcal{C}(\mathbf{m}_n(\mathbf{p})) - d_n|^2 = |\mathcal{C}(\mathbf{m}(t_n, \mathbf{p})) - d(t_n)|^2 + O(h),$$

uniformly in n . Multiplying by h and summing over n gives

$$J_{h,\infty}(\mathbf{p}) = J_\infty(\mathbf{p}) + O(h).$$

Thus, in the stable time-step range, the reduced discrete objective converges to the continuous reduced objective with first-order accuracy. The passage $J_\infty \rightarrow J_{h,\infty}$ therefore isolates the effect of time discretization before any penalty relaxation is introduced.

3.2 Continuous Penalty Relaxation

Quadratic penalty relaxations are a standard way to replace equality-constrained optimization problems by unconstrained or weakly constrained problems [10, 11]. Instead of enforcing (2.1) exactly, one may allow deviations from the dynamics and penalize them in the objective. This leads to the continuous penalized functional

$$J_\lambda(\mathbf{m}, \mathbf{p}) := \int_0^T |\mathcal{C}(\mathbf{m}(t)) - d(t)|^2 dt + \frac{\lambda}{2} \int_0^T \left\| \dot{\mathbf{m}}(t) - A(\mathbf{p}, t) \mathbf{m}(t) - \mathbf{q}(\mathbf{p}) \right\|_2^2 dt, \quad \lambda > 0. \quad (3.1)$$

For small λ the optimizer may deviate from the Bloch dynamic to improve the data fit; for large λ the residual term forces $\mathbf{m}$ to satisfy the Bloch equation increasingly well. This is the ODE analogue of the quadratic penalty formulations already introduced in Section 2.4, now stated at the continuous level. This continuous functional is included to make the approximation hierarchy explicit. The precise penalty estimate used in this thesis is stated at the discrete level in Section 3.4, where the constants in the $O(\lambda^{-1})$ term are discussed. In the numerical experiments, the penalty method is implemented after time discretization, so the fully discrete functional $J_{h,\lambda}$ is the main object of study.

3.3 Fully Discrete Penalized Formulation

The fully discrete penalized formulation is the object used in the penalty experiments. It is obtained by combining the Forward Euler discretization with the quadratic penalty relaxation introduced in Section 2.4. Thus the objective is $J_{h,\lambda}$, defined in (2.8), and the optimization variables are both the discrete trajectory $\mathbf{M}$ and the parameter vector $\mathbf{p}$.

This formulation differs from the reduced objective $J_{h,\infty}$. In the reduced formulation, each parameter vector determines its own trajectory through the Forward Euler recursion. In the penalized formulation, the trajectory is an additional optimization variable, and deviations from the discrete Bloch equation are allowed but penalized. Thus h controls the time discretization, while λ controls the strength of the model-consistency penalty.

3.4 Approximation Hierarchy and Consistency of the Limits

We now use the approximation hierarchy to compare the two limiting processes: refining the time discretization, $h \rightarrow 0$, and increasing the penalty parameter, $\lambda \rightarrow \infty$. The goal is to make precise when these two approximations are expected to lead to the same limiting reconstruction. We first recall the discretization estimate from Section 3.1. In the stable time-step range, and under the smoothness assumptions stated there, the reduced discrete objective satisfies

$$|J_{h,\infty}(\mathbf{p}) - J_\infty(\mathbf{p})| \leq K_{\text{disc}} h$$

for $\mathbf{p}$ in the admissible parameter set. The constant K_{disc} depends on the length of the time interval, the smoothness of the Bloch solution, the size of the magnetic field terms, the parameter bounds, and the stability constant of the Forward Euler method. This dependence is important: when h approaches the Forward Euler stability threshold, the asymptotic first-order behaviour may not be visible in practice. Next, consider the penalty approximation for fixed h . Let $(\bar{\mathbf{M}}_h, \bar{\mathbf{p}}_h)$ be a minimizer of the equality-constrained discrete problem. Since this pair satisfies the discrete ODE constraint, it is also admissible for the penalty formulation. Therefore,

$$J_{h,\lambda}^* \leq J_{h,\infty}^*,$$

where $J_{h,\lambda}^*$ and $J_{h,\infty}^*$ denote the corresponding minimum values. This inequality reflects the fact that the penalty formulation searches over a larger set of trajectories: the state trajectory is allowed to violate the discrete Bloch equation, but such violations are penalized. The reverse estimate is local. Assume that the constrained discrete problem is regular near $(\bar{\mathbf{M}}_h, \bar{\mathbf{p}}_h)$ and that a Lagrange multiplier $\boldsymbol{\eta}_h$ exists for the discrete ODE constraint. The first-order optimality conditions imply that, locally, any decrease of the data misfit below the constrained minimum must be accompanied by a violation of the constraint. The size of this possible decrease is controlled by the multiplier term. Applying Young's inequality to this term and to the quadratic penalty term gives the standard local quadratic-penalty estimate [10]

$$J_{h,\lambda}^* \geq J_{h,\infty}^* - \frac{K_{\text{pen}}(h)}{\lambda},$$

where $K_{\text{pen}}(h)$ depends on the multiplier norm and on the local conditioning of the discrete constraint. Consequently,

$$0 \leq J_{h,\infty}^* - J_{h,\lambda}^* \leq \frac{K_{\text{pen}}(h)}{\lambda}.$$

Combining the discretization and penalty estimates gives the objective-value bound

$$|J_{h,\lambda}^* - J_\infty^*| \leq K_{\text{disc}}h + \frac{K_{\text{pen}}(h)}{\lambda}.$$

For fixed h , this shows that

$$J_{h,\lambda}^* = J_{h,\infty}^* + O(\lambda^{-1}) \quad \text{as } \lambda \rightarrow \infty.$$

If $K_{\text{pen}}(h)$ remains bounded as $h \rightarrow 0$, then the combined estimate may be summarized formally as

$$J_{h,\lambda}^* = J_\infty^* + O(h) + O(\lambda^{-1}).$$

In this thesis, the dependence of $K_{\text{pen}}(h)$ on h is not analyzed theoretically. Therefore, the explicit bound involving $K_{\text{pen}}(h)$ will be used throughout. The estimate above concerns objective values, not directly parameter errors. In the noise-free synthetic setting, the data are generated from the true parameter $\mathbf{p}^*$, so $\mathbf{p}^*$ is the ideal minimizer of J_∞ . To translate objective-value errors into parameter errors, one needs a local stability condition. Suppose that J_∞ has local quadratic growth near $\mathbf{p}^*$, that is,

$$J_\infty(\mathbf{p}) \geq J_\infty(\mathbf{p}^*) + \frac{\alpha}{2} \|\mathbf{p} - \mathbf{p}^*\|^2$$

for $\mathbf{p}$ near $\mathbf{p}^*$, with $\alpha > 0$. If the parameter component $\mathbf{p}_{h,\lambda}$ of a penalized minimizer remains in this neighbourhood, then the objective-value estimate suggests the formal bound

$$\|\mathbf{p}_{h,\lambda} - \mathbf{p}^*\| \lesssim \left(\frac{2}{\alpha} \left[K_{\text{disc}}h + \frac{K_{\text{pen}}(h)}{\lambda} \right] \right)^{1/2}.$$

Thus, even if the objective values are close, the parameter error can remain large when the local curvature constant α is small. This is precisely the situation expected in poorly conditioned inverse problems and motivates the Hessian analysis in Chapter 4. The previous estimates concern objective values. The corresponding statement for minimizers requires additional stability assumptions, because convergence of objective values alone does not guarantee convergence of the reconstructed parameters. The following theorem formalizes the expected commutation of the limits for minimizers under such assumptions.

Theorem 3.1 (Consistency of minimizers under uniform approximation). *For each $h > 0$ and $\lambda > 0$, define the reduced penalized objective*

$$\Phi_{h,\lambda}(\mathbf{p}) = \inf_{\mathbf{M}} J_{h,\lambda}(\mathbf{M}, \mathbf{p}).$$

Thus a minimizer of $\Phi_{h,\lambda}$ is the parameter component of a minimizer of the fully discrete penalized problem. Assume that the admissible parameter set $\mathcal{P}$ is compact and that J_∞ is continuous on $\mathcal{P}$ with a unique minimizer $\mathbf{p}^*$. Assume further that the reduced discrete objectives converge uniformly to J_∞ , i.e.

$$\sup_{\mathbf{p} \in \mathcal{P}} |J_{h,\infty}(\mathbf{p}) - J_\infty(\mathbf{p})| \leq K_{\text{disc}} h,$$

and that the reduced penalized objectives converge uniformly to the reduced discrete objectives, i.e.

$$\sup_{\mathbf{p} \in \mathcal{P}} |\Phi_{h,\lambda}(\mathbf{p}) - J_{h,\infty}(\mathbf{p})| \leq \frac{K_{\text{pen}}(h)}{\lambda}.$$

Suppose also that $K_{\text{pen}}(h)$ remains bounded for sufficiently small stable time steps h , so that

$$K_{\text{pen}}(h) \leq K_{\text{pen}}.$$

Let $\widehat{\mathbf{p}}_{h,\lambda}$ be any minimizer of $\Phi_{h,\lambda}$ over $\mathcal{P}$. Then

$$\widehat{\mathbf{p}}_{h,\lambda} \longrightarrow \mathbf{p}^* \quad \text{as} \quad h \rightarrow 0, \quad \lambda \rightarrow \infty.$$

In particular, whenever the iterated limits exist,

$$\lim_{\lambda \rightarrow \infty} \lim_{h \rightarrow 0} \widehat{\mathbf{p}}_{h,\lambda} = \lim_{h \rightarrow 0} \lim_{\lambda \rightarrow \infty} \widehat{\mathbf{p}}_{h,\lambda} = \mathbf{p}^*.$$

Proof: The two uniform estimates imply that

$$\sup_{\mathbf{p} \in \mathcal{P}} |\Phi_{h,\lambda}(\mathbf{p}) - J_\infty(\mathbf{p})| \leq K_{\text{disc}} h + \frac{K_{\text{pen}}}{\lambda}.$$

Hence $\Phi_{h,\lambda}$ converges uniformly to J_∞ as $h \rightarrow 0$ and $\lambda \rightarrow \infty$. Let (h_k, λ_k) be any sequence such that

$$h_k \rightarrow 0, \quad \lambda_k \rightarrow \infty,$$

and let

$$\widehat{\mathbf{p}}_k = \widehat{\mathbf{p}}_{h_k, \lambda_k}$$

be a corresponding sequence of minimizers. Since $\mathcal{P}$ is compact, this sequence has a convergent subsequence. Denote this subsequence again by $\widehat{\mathbf{p}}_k$, and let

$$\widehat{\mathbf{p}}_k \rightarrow \bar{\mathbf{p}} \in \mathcal{P}.$$

Because $\widehat{\mathbf{p}}_k$ minimizes Φ_{h_k, λ_k} , we have

$$\Phi_{h_k, \lambda_k}(\widehat{\mathbf{p}}_k) \leq \Phi_{h_k, \lambda_k}(\mathbf{p}^*).$$

Using the uniform convergence of $\Phi_{h,\lambda}$ to J_∞ , it follows that

$$J_\infty(\widehat{\mathbf{p}}_k) \leq J_\infty(\mathbf{p}^*) + o(1).$$

Passing to the limit along the subsequence and using the continuity of J_∞ , we obtain

$$J_\infty(\bar{\mathbf{p}}) \leq J_\infty(\mathbf{p}^*).$$

Since $\mathbf{p}^*$ is the unique minimizer of J_∞ , this implies

$$\bar{\mathbf{p}} = \mathbf{p}^*.$$

Thus every convergent subsequence of $\hat{\mathbf{p}}_{h,\lambda}$ has the same limit $\mathbf{p}^*$. Since the admissible set is compact, this proves

$$\hat{\mathbf{p}}_{h,\lambda} \rightarrow \mathbf{p}^* \quad \text{as} \quad h \rightarrow 0, \quad \lambda \rightarrow \infty.$$

The statement about the iterated limits follows from the same joint convergence: whenever the inner limits exist, taking the remaining outer limit must give the same limit $\mathbf{p}^*$.

The assumptions in Theorem 3.1 are stronger than the objective-value estimate alone. They require uniform approximation of the relevant objectives and a unique stable minimizer of the ideal problem. This is important: objective values may converge even when minimizers do not, especially in inverse problems with flat or nearly flat directions. The theorem therefore gives the expected commutation of the limits only under stability assumptions. If the limiting objective has weak curvature, then the convergence of minimizers may be slow or unstable. This is precisely why the Hessian analysis in Chapter 4 is needed.

3.5 Numerical Experiments

All experiments are performed in a synthetic single-voxel setting. The true parameter vector is

$$\mathbf{p}^* = (T_1^*, T_2^*, \rho^*)^T = (2.0, 0.5, 1.1)^T,$$

and the observation window is $[0, 1]$. The constant effective field is chosen as

$$\mathbf{b} = (b_x, b_y, b_z)^T = (10, 10, 0)^T.$$

Unless stated otherwise, the data are noise-free and generated from a high-accuracy reference solution of the Bloch equation:

$$d_n = C(\mathbf{m}(t_n, \mathbf{p}^*), \mathbf{p}^*), \quad n = 0, \dots, N.$$

The initial magnetization is

$$\mathbf{m}_0 = (1, 0, 0)^T,$$

corresponding to a fully transverse initial state after an idealized 90° pulse. This choice gives a strong initial transverse signal and therefore provides a favourable setting for observing the influence of the relaxation parameters. The unknown parameter vector is reconstructed from the sampled data by minimizing the relevant discrete objective.

The ideal Gauss–Newton-type algorithms corresponding to the reduced and penalized formulations are given in Appendix 7. In the reduced formulation, the ideal algorithm updates only the parameter vector $\mathbf{p}$, while in the penalized formulation the ideal algorithm updates both the discrete trajectory $\mathbf{M}$ and the parameter vector $\mathbf{p}$. These algorithms describe the mathematical structure of the two least-squares problems. In the numerical experiments, however, the finite-dimensional optimization problems are solved using L-BFGS-B, a limited-memory quasi-Newton method for bound-constrained optimization [10]. This choice avoids explicitly assembling and solving the full Gauss–Newton systems and allows the parameter bounds to be imposed directly. The initial guess for the parameter vector is

$$\mathbf{p}^{(0)} = (1.5, 0.7, 1.0)^T.$$

The same initial guess is also used for the penalty experiments. The bound constraints are $T_1 \in [0.01, 5]$, $T_2 \in [0.01, 2]$, and $\rho \in [0.5, 1.5]$. The reconstruction error is measured by

$$\|\hat{\mathbf{p}} - \mathbf{p}^*\|.$$

3.5.1 Effect of the Time Step

The first experiment studies the reduced objective $J_{h,\infty}$ for different values of the time step h . For each value of h , the discrete trajectory is computed from the Forward Euler recursion for the current parameter vector, and the optimization is performed only over $\mathbf{p}$. The goal is to isolate the effect of the time discretization on the reconstructed parameters. Figure 3.1 shows the reconstruction error and the recovered parameter values as functions of h . The vertical dotted line marks the estimated critical Forward Euler time step h_{crit} , defined by the eigenvalue stability condition in (2.5). For the chosen true parameter vector and effective field, this gives

$$h_{\text{crit}} \approx 1.24 \cdot 10^{-2}.$$

For $h < h_{\text{crit}}$, the reconstruction error decreases as the time step is refined. A least-squares fit over this stable time-step range gives a slope of approximately 0.87 on the log-log scale. This is close to the first-order behaviour expected from Forward Euler, but it should not be interpreted as a pure ODE convergence rate. The quantity plotted is the error in the reconstructed parameter vector, so it also reflects the conditioning and sensitivity of the inverse problem. For $h > h_{\text{crit}}$, the error remains large because the discrete forward model is unstable or inaccurate. In this case, the inverse problem is effectively fitting the data with the wrong forward model, and the recovered parameters may be driven to the imposed bounds. These values should therefore not be interpreted physically. The right panel of Figure 3.1 shows that T_2 and ρ are already close to their true values for small and moderately small h , whereas T_1 converges more slowly. Thus, even within the stable time-step range, most of the remaining error comes from the T_1 component. This suggests that the remaining bias is not only a discretization effect, but is also related to the sensitivity structure of the inverse problem. This motivates

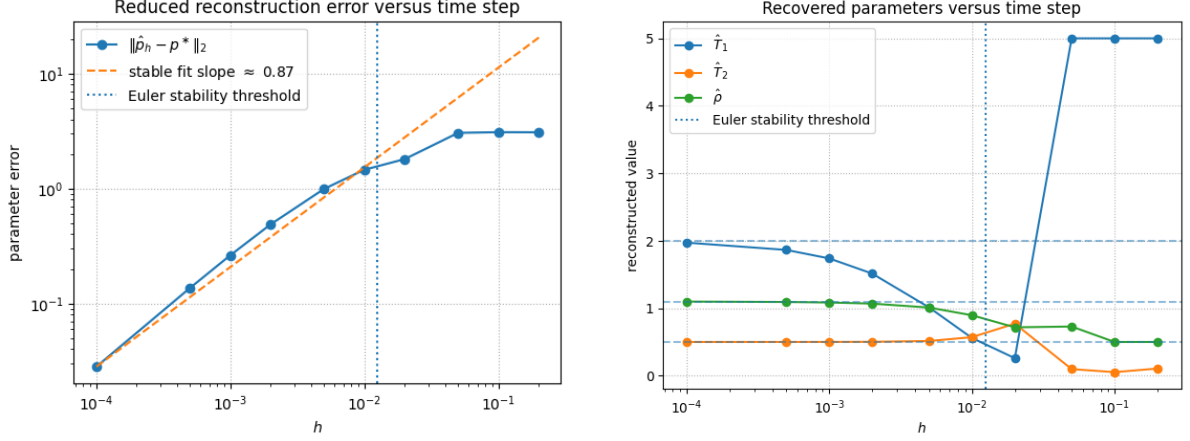


Figure 3.1: Left: reduced reconstruction error as a function of the time step h . Right: recovered parameters as functions of h . The vertical dotted line marks the estimated Forward Euler stability threshold h_{crit} , and dashed horizontal lines indicate the true parameter values.

the Hessian and identifiability analysis in Chapter 4. The fixed- h penalty experiment in the next subsection uses $\hat{h} = 10^{-3}$. This value lies well inside the stable time-step range and gives a reasonably accurate reduced reconstruction while remaining computationally inexpensive. In the joint experiment, h is varied again in order to study its interaction with λ .

3.5.2 Effect of the Penalty Parameter

The second experiment fixes the time step at $\hat{h} = 10^{-3}$ and studies the effect of the penalty parameter λ . For each value of λ , we minimize the fully discrete penalized objective $J_{\hat{h},\lambda}$ and record the data misfit D_h , the scaled ODE residual Ψ_h , and the parameter reconstruction error

$$\|\hat{\mathbf{p}}_{\hat{h},\lambda} - \mathbf{p}^*\|.$$

The quantities D_h and Ψ_h are the terms defined in (2.6) and (2.7).

λ	parameter error	data misfit	scaled ODE residual
1	0.566	0.808	$1.176 \cdot 10^{-1}$
10	0.601	1.510	$2.354 \cdot 10^{-3}$
10^2	0.605	1.627	$1.174 \cdot 10^{-4}$
10^3	0.590	1.658	$4.881 \cdot 10^{-5}$
10^4	0.560	1.793	$3.956 \cdot 10^{-6}$

Table 3.1: Penalty diagnostics at $\hat{h} = 10^{-3}$.

Figure 3.2 shows the L-curve, a standard diagnostic for visualizing the trade-off be-

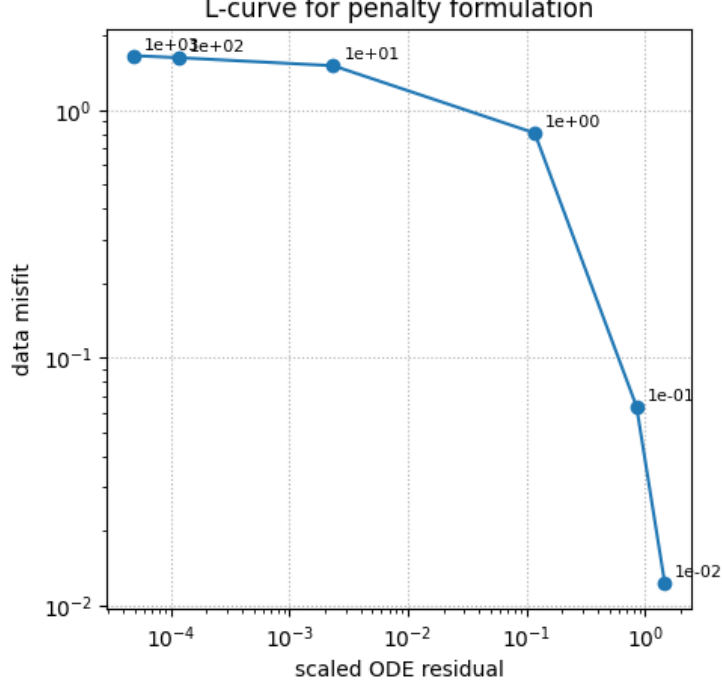


Figure 3.2: L-curve for the fully discrete penalty formulation at $\hat{h} = 10^{-3}$. The horizontal axis shows the scaled ODE residual Ψ_h , while the vertical axis shows the data misfit D_h . Each point corresponds to one value of λ . Small values of λ give low data misfit but larger model residual, while large values of λ enforce the discrete Bloch dynamics more strongly at the cost of a larger data misfit.

tween competing terms in inverse problems [13, 14]. The data misfit is plotted against the scaled ODE residual as λ varies. For small values of λ , the penalty on the ODE residual is weak, so the optimization can fit the data well while allowing larger violations of the discrete Bloch equation. For large values of λ , violations of the discrete dynamics become expensive, and the scaled ODE residual decreases. This improved model consistency is obtained at the cost of a larger data misfit. The diagnostic values in Table 3.1 show the same effect quantitatively. As λ increases, the scaled ODE residual decreases by several orders of magnitude, while the data misfit increases. However, the parameter error does not decrease monotonically. This shows that enforcing the discrete Bloch equation more strongly is not, by itself, sufficient to guarantee a better parameter reconstruction. This observation is important for the interpretation of the penalty method. The penalty parameter controls the balance between fitting the measured signal and satisfying the discrete Bloch dynamics, but the quality of the recovered parameters also depends on the sensitivity structure of the inverse problem and on the convergence of the numerical optimization. The remaining reconstruction error, especially in the T_1 component, therefore motivates the identifiability analysis in Chapter 4.

3.5.3 Joint Dependence on h and λ

The previous experiments studied h and λ separately. We now examine their joint influence by solving the penalty problem for several time steps and penalty parameters. This experiment is intended as a finite-dimensional numerical analogue of the consistency discussion in Section 3.4: it tests how the reconstruction changes when the time discretization is refined and the ODE residual is penalized more strongly.

h	reduced error	$\lambda = 1$	$\lambda = 10$	$\lambda = 10^2$	$\lambda = 10^3$	$\lambda = 10^4$
10^{-2}	1.453	0.734	0.914	0.855	0.695	0.573
$5 \cdot 10^{-3}$	0.990	0.676	0.729	0.693	0.639	0.566
$2 \cdot 10^{-3}$	0.488	0.602	0.624	0.624	0.598	0.559
10^{-3}	0.262	0.566	0.601	0.605	0.590	0.560

Table 3.2: Reconstruction errors for the reduced and penalized formulations for several time steps h and penalty parameters λ . The reduced error is $\|\hat{\mathbf{p}}_h - \mathbf{p}^*\|_2$. The remaining columns report $\|\hat{\mathbf{p}}_{h,\lambda} - \mathbf{p}^*\|_2$ for the penalty formulation.

Table 3.2 reports the parameter reconstruction errors for the reduced and penalized formulations. The reduced formulation is strongly affected by the time step. As h decreases from 10^{-2} to 10^{-3} , the reduced reconstruction error decreases from approximately 1.45 to 0.26. This agrees with the time-step study in Figure 3.1: once the Forward Euler scheme is stable, refining h improves the reduced reconstruction. The penalty formulation behaves differently. For the coarser time steps $h = 10^{-2}$ and $h = 5 \cdot 10^{-3}$, the penalty formulation gives smaller parameter errors than the reduced formulation. This suggests that allowing a controlled violation of the discrete Bloch equation can partially compensate for discretization error in the forward model. However, for the finer time steps $h = 2 \cdot 10^{-3}$ and $h = 10^{-3}$, the reduced formulation becomes more accurate than the penalized reconstructions in the tested range of λ . The detailed penalty diagnostics show the expected trade-off between the data misfit and the ODE residual. For each fixed h , increasing λ reduces the scaled ODE residual by several orders of magnitude, while the data misfit increases. Thus, larger λ enforces the discrete dynamics more strongly, but this does not automatically lead to a smaller parameter error. In particular, for $h = 10^{-3}$, the reduced error is approximately 0.26, whereas the penalized errors remain around 0.56 or larger for the tested values of λ . This observation is important for interpreting the commutation question. In theory, an exact minimizer of the quadratic penalty problem should approach the constrained discrete solution as $\lambda \rightarrow \infty$. In the present computations, however, the penalty problem is solved by a finite number of alternating updates and is therefore also affected by algorithmic error and possible local minima. The discrepancy between the large- λ penalty solutions and the reduced solution should therefore not be interpreted as a failure of the formal penalty principle alone. It reflects the combined effect of finite λ , finite iteration count, nonconvexity, and the conditioning of the inverse problem. Taken together, the experiments show that both discretization and penalty enforcement have a substantial influence on

parameter recovery. Refining h improves the reduced formulation, while increasing λ improves model consistency but may increase the data misfit and does not necessarily improve the parameter estimate. The remaining difficulty, especially in the recovery of T_1 , motivates the sensitivity and identifiability analysis in Chapter 4.

4 Sensitivity Analysis and Identifiability

The numerical experiments in Chapter 3 showed that refining the time step improves the reduced reconstruction once the Forward Euler discretization is in the stable time-step range. However, the reconstruction error does not disappear completely. In particular, the recovery of T_1 remains weaker than the recovery of T_2 and ρ . This suggests that the remaining error is not only caused by time discretization or penalty enforcement, but also by the local sensitivity structure of the inverse problem. The purpose of this chapter is to analyze this sensitivity structure. We study the local geometry of the reduced data-misfit functional using contour plots, the Jacobian of the residual, and the Gauss–Newton Hessian. The eigenvalues and eigenvectors of this Hessian indicate which parameter combinations are strongly constrained by the data and which directions remain weakly identifiable. This provides a local explanation for the reconstruction behaviour observed in Chapter 3.

4.1 Visual diagnosis: contour plots of the reduced objective

We first visualize the reduced objective in two-dimensional parameter planes. In each plot, two parameters are varied while the third parameter is fixed at its true value. The true parameter pair is marked by a red point. The objective values are computed using the same single-voxel setup as in the Hessian analysis below. Figure 4.1 gives a geometric first indication of which parameter directions are well identified. In the two planes involving T_1 , the low-objective region is elongated in the T_1 -direction. This means that changes in T_1 have a relatively small effect on the data misfit. In contrast, the (T_2, ρ) -plane shows more curved contours, indicating stronger sensitivity to these parameters. The tilt of these contours suggests coupling between transverse relaxation and signal amplitude. These observations are only visual diagnostics. The Hessian analysis below quantifies them by identifying the weakest local direction and by measuring the parameter coupling through the normalized Hessian correlation matrix.

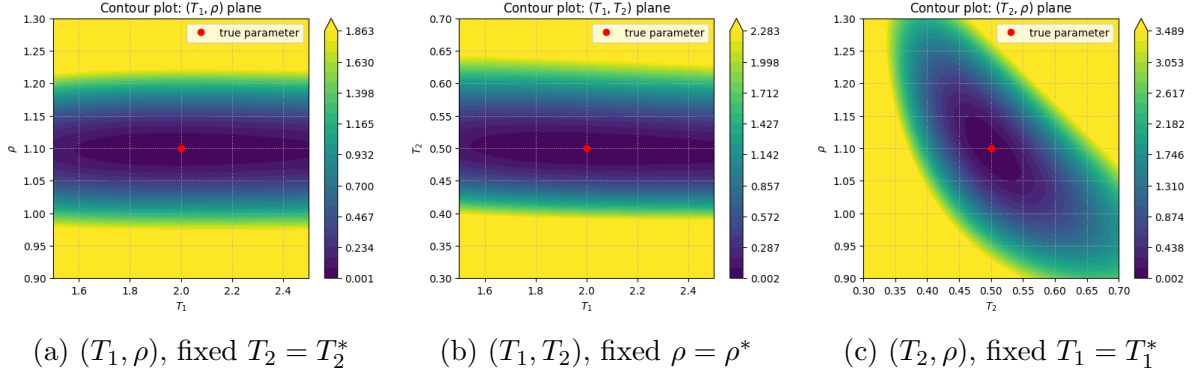


Figure 4.1: Contour plots of the reduced objective in two-dimensional parameter planes through the true parameter vector $\mathbf{p}^*$.

4.2 Local quadratic model

Recall that the reduced discrete data-misfit functional is

$$\Phi_h(\mathbf{p}) = \frac{h}{2} \sum_{n=0}^N |\mathcal{C}(\mathbf{m}_n(\mathbf{p}), \mathbf{p}) - d_n|^2,$$

where $\mathbf{m}_n(\mathbf{p})$ denotes the discrete magnetization trajectory obtained from the Forward Euler method. The measurement operator is

$$\mathcal{C}(\mathbf{m}, \mathbf{p}) = \rho C \mathbf{m}, \quad C = (1 \ i \ 0).$$

Since the measured signal is complex-valued, we represent the residual in real form by stacking real and imaginary parts. For

$$\tilde{r}_n(\mathbf{p}) = \mathcal{C}(\mathbf{m}_n(\mathbf{p}), \mathbf{p}) - d_n \in \mathbb{C},$$

we define

$$\mathbf{r}(\mathbf{p}) = \begin{pmatrix} \operatorname{Re} \tilde{r}_0(\mathbf{p}) \\ \operatorname{Im} \tilde{r}_0(\mathbf{p}) \\ \vdots \\ \operatorname{Re} \tilde{r}_N(\mathbf{p}) \\ \operatorname{Im} \tilde{r}_N(\mathbf{p}) \end{pmatrix} \in \mathbb{R}^{2(N+1)}.$$

Then the reduced objective can be written as

$$\Phi_h(\mathbf{p}) = \frac{h}{2} \|\mathbf{r}(\mathbf{p})\|^2.$$

Let

$$J_r(\mathbf{p}) = \frac{\partial \mathbf{r}(\mathbf{p})}{\partial \mathbf{p}}$$

denote the real residual Jacobian. For a perturbation $\delta \mathbf{p}$, the first variation is

$$\delta \Phi_h(\mathbf{p})[\delta \mathbf{p}] = h \langle \mathbf{r}(\mathbf{p}), J_r(\mathbf{p}) \delta \mathbf{p} \rangle,$$

and hence

$$\nabla \Phi_h(\mathbf{p}) = h J_r(\mathbf{p})^T \mathbf{r}(\mathbf{p}).$$

Differentiating once more gives

$$\nabla^2 \Phi_h(\mathbf{p}) = h J_r(\mathbf{p})^T J_r(\mathbf{p}) + h \sum_{k=1}^{2(N+1)} r_k(\mathbf{p}) \nabla^2 r_k(\mathbf{p}).$$

For noise-free data and sufficiently accurate time discretization, the residual near $\mathbf{p}^*$ is small. The Gauss–Newton approximation is standard for nonlinear least-squares problems and neglects the second-order residual terms when the residual is small [10, 9]. In this thesis, we use

$$H(\mathbf{p}) = J_r(\mathbf{p})^T J_r(\mathbf{p})$$

as the Gauss–Newton Hessian diagnostic. The omitted factor h only rescales all eigenvalues by the same positive constant and therefore does not change the eigenvectors, relative sensitivities, or condition number. Thus it does not affect the identifiability directions discussed below. The remainder of this chapter uses $H(\mathbf{p})$ as a local diagnostic for identifiability. Its eigenvalues and eigenvectors reveal which parameter combinations are well constrained by the data and which remain weakly observable.

4.3 Spectral analysis and conditioning

Let

$$H = Q \operatorname{diag}(\mu_1, \mu_2, \mu_3) Q^T, \quad \mu_1 \geq \mu_2 \geq \mu_3 \geq 0,$$

be the spectral decomposition of the symmetric positive semi-definite matrix H . The columns of $Q = (\mathbf{q}_1, \mathbf{q}_2, \mathbf{q}_3)$ are orthonormal eigenvectors. The local quadratic approximation around the true parameter vector is

$$\Phi_h(\mathbf{p}^* + \delta \mathbf{p}) \approx \Phi_h(\mathbf{p}^*) + \frac{1}{2} \sum_{i=1}^3 \mu_i (\mathbf{q}_i^T \delta \mathbf{p})^2.$$

Large eigenvalues correspond to directions in which the objective increases rapidly, so the corresponding parameter combinations are strongly constrained by the data. Small eigenvalues correspond to nearly flat directions, where changes in the parameters have little effect on the measured signal. The condition number

$$\kappa(H) = \frac{\mu_1}{\mu_3}$$

quantifies this anisotropy. Here, anisotropy means that the objective does not curve equally in all parameter directions: in some directions it increases rapidly, while in

others it remains relatively flat. The eigenvalues of the local Hessian provide a standard diagnostic for local conditioning and parameter identifiability in inverse problems [7, 6]. When μ_3 is small, the Hessian is nearly singular and the inverse problem is locally ill-conditioned. In that case, small perturbations in the data can produce large changes in the reconstructed parameter vector. Geometrically, the level sets of the quadratic model are ellipsoids. A large condition number produces highly elongated ellipsoids, with steep directions and nearly flat directions. In the context of inverse problems, the smallest eigenvalues are often the most important because they indicate directions in parameter space that are only weakly visible to the measurements.

4.4 Parameter sensitivities and identifiability

The local identifiability of $\mathbf{p}$ is determined by how the predicted signal changes when the parameters are perturbed. Define the state sensitivity matrix at time t_n by

$$S_n(\mathbf{p}) = \frac{\partial \mathbf{m}(t_n, \mathbf{p})}{\partial \mathbf{p}} \in \mathbb{R}^{3 \times 3}.$$

After applying the measurement operator and stacking the real and imaginary parts, one obtains the residual Jacobian $J_r(\mathbf{p})$. The columns of $J_r(\mathbf{p})$ measure how strongly the predicted data change when T_1 , T_2 , or ρ is perturbed. The Gauss–Newton Hessian entries satisfy

$$H_{ij}(\mathbf{p}) = \langle J_{r,i}(\mathbf{p}), J_{r,j}(\mathbf{p}) \rangle, \quad H_{ii}(\mathbf{p}) = \|J_{r,i}(\mathbf{p})\|^2.$$

Sensitivity matrices and Hessian-based diagnostics are commonly used to assess local parameter identifiability and uncertainty in inverse problems [7, 15]. The quantities $\sqrt{H_{ii}}$ are therefore local sensitivity norms of the measured signal with respect to the individual parameters. They measure the size of the change in the predicted data caused by a unit perturbation of T_1 , T_2 , or ρ . Since the parameters have different physical scales, these values should be interpreted as local diagnostics for the chosen parameterization and bounds, rather than as universal physical constants. Large diagonal entries indicate strongly constrained parameters; small diagonal entries indicate weak sensitivity. Large off-diagonal entries indicate that two parameters affect the data in similar directions, making their individual effects more difficult to separate.

4.5 Hessian structure and parameter coupling

We now evaluate the Gauss–Newton Hessian for the reduced problem at the true parameter vector

$$\mathbf{p}^* = (2.0, 0.5, 1.1)^T.$$

The computation uses the stable time step

$$h_{\text{loc}} = 10^{-4}.$$

The data are generated from the high-accuracy reference solution, and the complex residual is represented by stacking real and imaginary parts. The Jacobian is computed by central finite differences. The resulting Hessian is

$$H = \begin{pmatrix} 3.16 \cdot 10^{-1} & 1.77 & 7.57 \cdot 10^{-2} \\ 1.77 & 1.55 \cdot 10^2 & 9.82 \cdot 10^1 \\ 7.57 \cdot 10^{-2} & 9.82 \cdot 10^1 & 1.20 \cdot 10^2 \end{pmatrix}.$$

A finite-difference robustness check was performed using relative steps from 10^{-3} to 10^{-7} . The eigenvalues were unchanged to the displayed precision, indicating that the Hessian diagnostics are not an artefact of the finite-difference step. The sensitivity norms are

$$\sqrt{\text{diag}(H)} = (0.562, 12.452, 10.962),$$

corresponding to T_1 , T_2 , and ρ , respectively. Normalizing by the largest value shows that the relative sensitivity of T_1 is only about 5% of the sensitivity of T_2 . This provides a local explanation for the dominant T_1 error observed in Chapter 3.

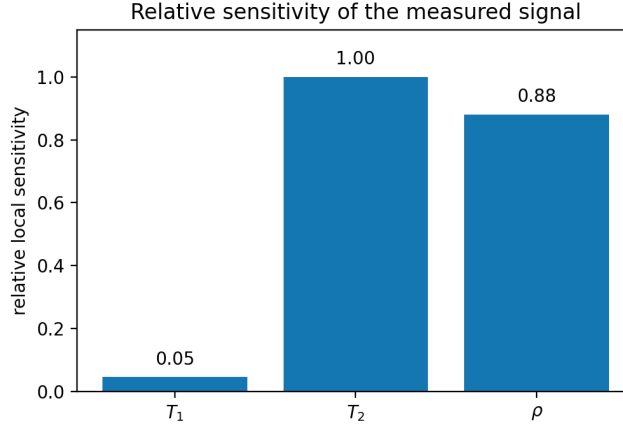


Figure 4.2: Relative local sensitivity of the measured signal with respect to T_1 , T_2 , and ρ , computed from the diagonal entries of the Gauss–Newton Hessian and normalized by the largest value.

The eigenvalues of H are

$$\mu_1 = 237.37, \quad \mu_2 = 37.89, \quad \mu_3 = 0.277,$$

giving $\kappa(H) \approx 856$. The weakest eigenvector is

$$\mathbf{q}_3 = (0.9996, -0.0229, 0.0181)^T.$$

Thus the flattest local direction is almost aligned with the T_1 -axis. The problem is not singular, but it is strongly anisotropic: perturbations in T_1 produce much smaller changes

in the measured signal than comparable perturbations in T_2 or ρ . The normalized Hessian correlation matrix is

$$\text{corr}(H) = \begin{pmatrix} 1 & 0.25 & 0.01 \\ 0.25 & 1 & 0.72 \\ 0.01 & 0.72 & 1 \end{pmatrix}.$$

The strongest off-diagonal correlation is between T_2 and ρ , while the correlation between T_1 and ρ is close to zero. Thus, in this experiment, the main difficulty with T_1 is weak sensitivity rather than strong coupling with ρ .

4.6 Relation to the penalty formulation

The Hessian analysis above is performed for the reduced formulation, where the state trajectory is determined by the Forward Euler recursion and the objective depends only on $\mathbf{p}$. The penalty formulation has a different local curvature because it treats the full trajectory $\mathbf{M}$ and the parameter vector $\mathbf{p}$ as optimization variables. Its all-at-once Hessian therefore contains state-state, state-parameter, and parameter-parameter blocks. A fully reduced curvature for the penalty problem would require eliminating the state variables, for example through a Schur complement with respect to the state block [9, 10]. This is not the diagnostic used here. Instead, we report only the parameter block of the penalty Gauss–Newton Hessian as an additional local diagnostic. This diagnostic is motivated by the penalty formulation for inverse problems discussed in [11]. Table 4.1 reports the eigenvalues and condition numbers of this parameter block for several values of λ .

λ	$\mu_{\max}$	μ_{mid}	$\mu_{\min}$	κ
1	$3.712 \cdot 10^0$	$5.202 \cdot 10^{-1}$	$3.646 \cdot 10^{-2}$	$1.018 \cdot 10^2$
10	$3.712 \cdot 10^1$	$3.379 \cdot 10^0$	$9.998 \cdot 10^{-2}$	$3.713 \cdot 10^2$
10^2	$3.712 \cdot 10^2$	$3.213 \cdot 10^1$	$5.661 \cdot 10^{-1}$	$6.558 \cdot 10^2$
10^3	$3.712 \cdot 10^3$	$3.197 \cdot 10^2$	$5.202 \cdot 10^0$	$7.136 \cdot 10^2$
10^4	$3.712 \cdot 10^4$	$3.195 \cdot 10^3$	$5.156 \cdot 10^1$	$7.200 \cdot 10^2$

Table 4.1: Eigenvalues and condition numbers of the parameter block of the Gauss–Newton Hessian for the penalty formulation.

Increasing λ increases the overall curvature associated with the ODE residual, but it does not improve the parameter-block condition number. Instead, the condition number increases and then plateaus. This is consistent with the observation in Chapter 3 that stronger enforcement of the discrete Bloch equation does not automatically improve parameter recovery. Thus, the penalty parameter controls model consistency, but conditioning and identifiability are governed by the local sensitivity structure of the measured signal. Even when the ODE residual is small, the measured transverse signal can remain weakly sensitive to T_1 .

4.7 Interpretation of the sensitivity structure

The weak T_1 sensitivity can be understood from the measurement model and from the chosen initial condition. The single-voxel measurement operator is

$$\mathcal{C}(\mathbf{m}, \mathbf{p}) = \rho(m_x + im_y),$$

so the measured signal directly observes only the transverse components of the magnetization. The parameter T_2 appears directly in the decay rates of the transverse equations, and therefore changes the observed signal strongly. The parameter ρ appears as a multiplicative factor in the measurement operator, so it also has a direct influence on the signal amplitude. By contrast, T_1 controls relaxation in the longitudinal component m_z and the source term ρ/T_1 . Since m_z is not measured directly, information about T_1 reaches the observed signal only indirectly through the coupling of the Bloch equations. For the chosen initial condition $\mathbf{m}_0 = (1, 0, 0)^T$ and observation window $[0, 1]$, this indirect influence is much weaker than the direct effects of T_2 and ρ . This explains why the sensitivity norm for T_1 is only about 5% of the sensitivity norm for T_2 . The coupling between T_2 and ρ has a different origin. Both parameters affect the transverse signal amplitude: T_2 changes the decay of the transverse magnetization, while ρ rescales the measured signal. Over a finite observation window, a change in T_2 can therefore be partially compensated by a change in ρ , producing the tilted contours in the (T_2, ρ) -plane and the correlation value 0.72 in the Hessian correlation matrix.

4.8 Conclusion

The local sensitivity analysis explains the reconstruction behaviour observed in Chapter 3. The Gauss–Newton Hessian has condition number approximately 856, showing that the reduced inverse problem is locally anisotropic. The weakest eigenvector is almost aligned with the T_1 -axis, and the relative sensitivity of the signal to T_1 is only about 5% of the sensitivity to T_2 . This identifies weak T_1 sensitivity as the main source of the remaining reconstruction error. At the same time, the normalized Hessian correlation matrix shows strong coupling between T_2 and ρ . The contour plots confirm both effects geometrically: planes involving T_1 contain elongated low-objective regions, while the (T_2, ρ) -plane shows tilted contours. Therefore, the remaining error in the reconstruction is not only a numerical discretization effect, but also a consequence of the intrinsic local geometry of the Bloch inverse problem.

5 Noise Sensitivity

The previous chapters studied the effects of time discretization, penalty enforcement, and local identifiability using noise-free data. In practice, MRI measurements are affected by measurement noise. This chapter investigates how additive complex-valued noise affects the reduced reconstruction in the chosen Bloch inverse problem. The goal is to distinguish between deterministic reconstruction error and noise-induced variability. The deterministic component is already present in the noise-free reconstruction and is caused by the finite time step together with the weak sensitivity of the data to T_1 . The stochastic component is introduced by noise in the measured complex signal. The results show that, for the tested noise levels, the mean reconstruction error remains close to the zero-noise baseline, while the variability of the recovered parameters increases with the noise level, especially in the T_1 component.

5.1 Noise model

Let

$$\mathbf{p}^* = (T_1^*, T_2^*, \rho^*)^T = (2.0, 0.5, 1.1)^T$$

denote the true parameter vector. The noise-free complex data are generated from the reference Bloch trajectory:

$$d_n = \mathcal{C}(\mathbf{m}(t_n, \mathbf{p}^*), \mathbf{p}^*), \quad n = 0, \dots, N.$$

Since the measured MRI signal is complex-valued, noise is added to both the real and imaginary parts. The noisy data are

$$d_n^\sigma = d_n + \eta_n,$$

where

$$\eta_n = \sigma \frac{\|d\|_2}{\sqrt{2(N+1)}} (\xi_n + i\zeta_n), \quad \xi_n, \zeta_n \sim \mathcal{N}(0, 1).$$

The random variables ξ_n and ζ_n are independent for all measurement times. With this scaling,

$$\mathbb{E}\|\eta\|_2^2 = \sigma^2 \|d\|_2^2,$$

so σ represents a relative noise level. Thus $\sigma = 10^{-2}$ corresponds approximately to one percent relative noise in the complex data. Figure 5.1 shows the magnitude of the clean and noisy measured signals for two representative noise levels. The displayed realized relative noise norms are close to the prescribed values of σ , confirming that the numerical noise scaling behaves as intended.

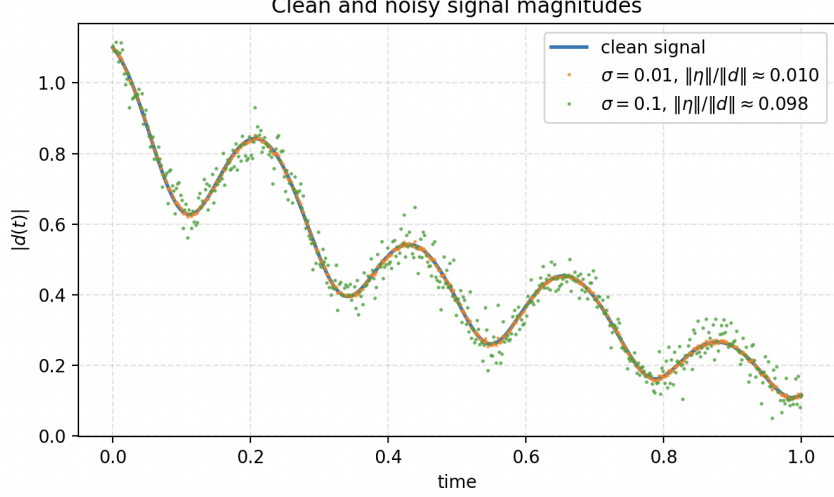


Figure 5.1: Magnitude of the clean and noisy measured signals for two representative relative noise levels.

5.2 Noise sensitivity of the reduced reconstruction

Noise sensitivity is especially important in inverse problems, where ill-conditioning can amplify perturbations in the measured data [6, 7]. In the present setting, the Hessian analysis of Chapter 4 suggests that such amplification should be strongest in the weak T_1 direction. The reduced reconstruction is performed with the same model and parameter values as in the previous chapters. No penalty parameter is used in this chapter, since the noise experiment is carried out for the reduced formulation. For each relative noise level

$$\sigma \in \{0, 10^{-4}, 10^{-3}, 5 \cdot 10^{-3}, 10^{-2}, 5 \cdot 10^{-2}, 10^{-1}\},$$

we perform 50 independent reconstructions from the same initial guess. The reconstruction error is measured by

$$\|\hat{\mathbf{p}}^\sigma - \mathbf{p}^*\|.$$

Figure 5.2 shows the mean reconstruction error over the noise realizations. The dashed horizontal line indicates the zero-noise baseline. Over the tested range, the mean error remains close to this baseline, while the error bars grow with σ . Thus, the main visible effect of noise is not a systematic shift of the mean reconstruction error, but an increase in variability.

5.3 Parameter-wise variability

The total reconstruction error does not show which parameter is most affected by noise. For this reason, we also examine the standard deviation of each reconstructed parameter component. Figure 5.3 shows the parameter-wise standard deviations as functions of the relative noise level. The variability grows approximately proportionally to σ , which is

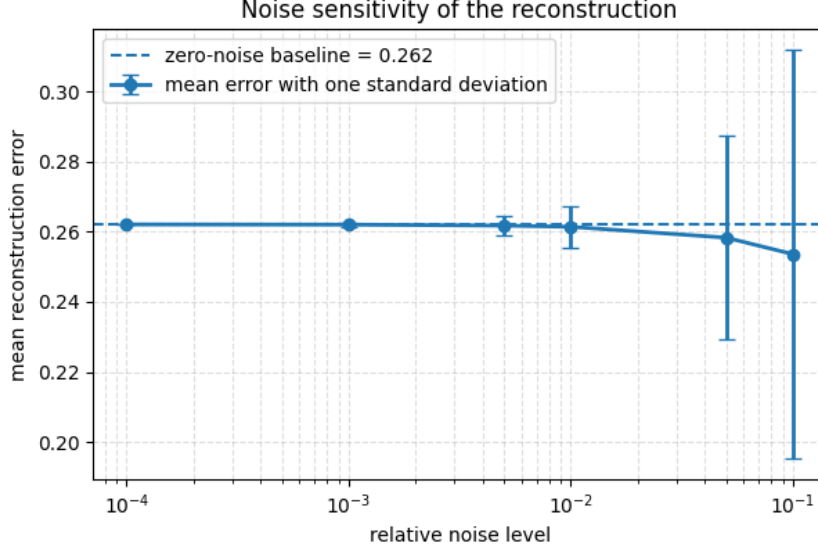


Figure 5.2: Noise sensitivity of the reduced reconstruction. Markers show the mean reconstruction error over 50 independent noise realizations, and error bars show one standard deviation. The dashed line indicates the zero-noise baseline.

the expected behaviour for a locally linear inverse problem. However, the size of the variability is very different for the three parameters. The standard deviation of $\hat{T}_1$ is much larger than the standard deviations of $\hat{T}_2$ and $\hat{\rho}$ across all tested noise levels. At $\sigma = 10^{-1}$, for example,

$$\text{std}(\hat{T}_1) \approx 5.84 \cdot 10^{-2},$$

whereas

$$\text{std}(\hat{T}_2) \approx 3.90 \cdot 10^{-3}, \quad \text{std}(\hat{\rho}) \approx 4.03 \cdot 10^{-3}.$$

Thus the noise-induced variability is dominated by the same parameter direction that was identified as weakly sensitive in Chapter ??.

5.4 Interpretation

The noise experiment separates two effects. First, there is a deterministic error already present at $\sigma = 0$. This zero-noise error is approximately

$$2.62 \cdot 10^{-1},$$

and is almost entirely due to the underestimation of T_1 . This deterministic component is caused by the finite time step together with the weak sensitivity of the measured transverse signal to T_1 . Second, as the relative noise level increases, the reconstruction becomes more variable. The mean error does not increase monotonically over the tested range, but this should not be interpreted as an improvement caused by noise. The zero-noise reconstruction underestimates T_1 , and random perturbations spread the estimates

σ	mean error	std. error	$T_1^* - \widehat{\widehat{T}}_1$	std. ($\widehat{T}_1$)
0	$2.621 \cdot 10^{-1}$	0	$2.616 \cdot 10^{-1}$	0
10^{-4}	$2.621 \cdot 10^{-1}$	$5.842 \cdot 10^{-5}$	$2.616 \cdot 10^{-1}$	$5.844 \cdot 10^{-5}$
10^{-3}	$2.620 \cdot 10^{-1}$	$5.843 \cdot 10^{-4}$	$2.616 \cdot 10^{-1}$	$5.845 \cdot 10^{-4}$
$5 \cdot 10^{-3}$	$2.617 \cdot 10^{-1}$	$2.921 \cdot 10^{-3}$	$2.613 \cdot 10^{-1}$	$2.922 \cdot 10^{-3}$
10^{-2}	$2.614 \cdot 10^{-1}$	$5.840 \cdot 10^{-3}$	$2.609 \cdot 10^{-1}$	$5.842 \cdot 10^{-3}$
$5 \cdot 10^{-2}$	$2.583 \cdot 10^{-1}$	$2.916 \cdot 10^{-2}$	$2.578 \cdot 10^{-1}$	$2.918 \cdot 10^{-2}$
10^{-1}	$2.536 \cdot 10^{-1}$	$5.829 \cdot 10^{-2}$	$2.530 \cdot 10^{-1}$	$5.836 \cdot 10^{-2}$

Table 5.1: Noise sensitivity of the reduced reconstruction with relative complex-valued noise.

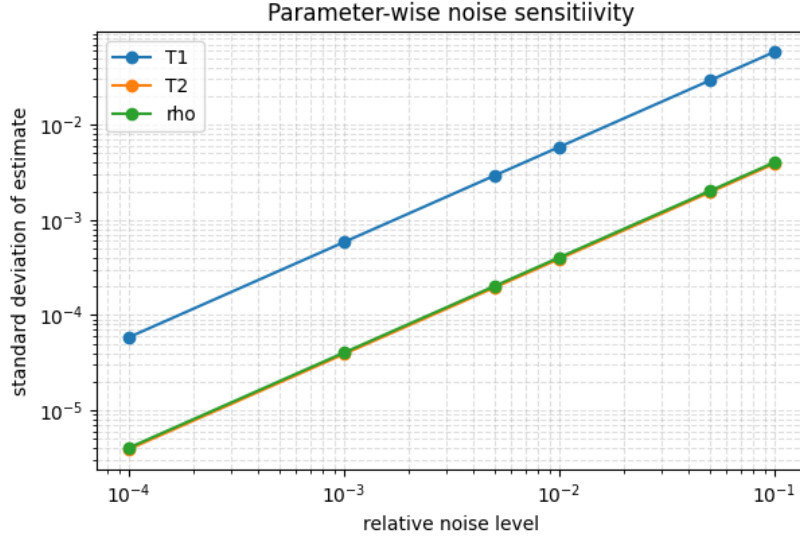


Figure 5.3: Parameter-wise standard deviation of the reconstructed parameters as a function of the relative noise level.

primarily along the weak T_1 direction. Some noise realizations therefore move the estimate closer to T_1^* , which can slightly reduce the sample mean error. The increasing standard deviation is the more informative indicator of noise sensitivity. The table confirms that the mean error remains close to the zero-noise baseline, while the standard deviation grows with σ . The last two columns show that both the dominant deterministic bias and the dominant noise-induced variability occur in the T_1 component. This behaviour is consistent with the local Hessian analysis. In Chapter 4, the weakest eigenvector of the Gauss–Newton Hessian was almost aligned with the T_1 -axis. Therefore, noise is expected to be amplified most strongly in the T_1 direction. The parameter-wise standard deviations in Figure 5.3 confirm this prediction.

5.5 Conclusion

The relative-noise experiments show that the reduced reconstruction is dominated by a deterministic T_1 error at the chosen time step. Adding complex-valued measurement noise does not substantially shift the mean reconstruction error over the tested range, but it increases the uncertainty of the recovered parameters. This uncertainty is highly anisotropic: the T_1 component varies much more strongly than T_2 and ρ . Together with the results of Chapters 3 and 4, this suggests the following interpretation. The reconstruction error contains a deterministic component associated with time discretization and weak local sensitivity, and a stochastic component associated with measurement noise:

$$\text{error} \approx \text{deterministic bias} + \text{noise-induced variability}.$$

For the penalty formulation studied in Chapter 3, an additional model-consistency contribution controlled by λ enters the same balance. Accurate parameter recovery therefore requires not only small h and appropriate penalty enforcement, but also an experimental setup in which the measured signal is sufficiently sensitive to all parameters, especially T_1 .

6 Discussion

This thesis studied a simplified ODE-constrained inverse problem motivated by quantitative MRI. The goal was to reconstruct the parameter vector $\mathbf{p} = (T_1, T_2, \rho)^T$ from complex-valued measurements of the transverse magnetization in a single-voxel Bloch model. The main focus was to understand how time discretization, penalty inclusion and local parameter sensitivity influence the reconstruction. The numerical experiments show that these effects cannot be interpreted independently. The Forward Euler time step h controls the accuracy and stability of the discrete forward model. When h is too large, the discrete dynamics no longer approximate the continuous Bloch trajectory reliably, and the inverse problem effectively fits the data with an inaccurate forward model. Once the time step lies in the stable time-step range, refining h improves the reduced reconstruction. However, even for small stable time steps, the reconstruction error does not disappear completely. This indicates that the remaining error is not only a discretization effect. The penalty parameter λ controls a different aspect of the computation. In the penalized formulation, the state trajectory is allowed to deviate from the discrete Bloch equation, but such deviations are penalized. Increasing λ reduces the discrete ODE residual and therefore improves model consistency. However, the experiments show that stronger model enforcement does not automatically imply better parameter recovery. In particular, the parameter error is not monotone in λ . For some coarser time steps, allowing a controlled violation of the discrete model can improve the parameter estimate compared with the reduced formulation. For finer time steps, the reduced formulation becomes more accurate than the penalized reconstructions in the tested range of λ . This illustrates that the best reconstruction is determined by a balance between data fit, model consistency, discretization error, and optimization error. The approximation hierarchy in Chapter 3 gives a formal way to think about the limits $h \rightarrow 0$ and $\lambda \rightarrow \infty$. At the level of objective values, one expects consistency when the time-discretization error and the penalty error are both small. However, the numerical results show why this formal statement must be interpreted carefully. First, the constants in the estimates matter. If the time step is close to the stability threshold, the discretization error may be large even if the formal order is first order. Second, objective-value closeness does not directly imply parameter closeness unless the inverse problem has enough local curvature. If the Hessian has a small eigenvalue, a small change in the objective can still correspond to a relatively large change in the reconstructed parameter vector. This point is confirmed by the sensitivity analysis in Chapter ???. The Gauss–Newton Hessian shows that the reduced inverse problem is locally anisotropic. The weakest eigenvector is almost aligned with the T_1 -axis, and the sensitivity of the measured signal to T_1 is much smaller than the sensitivities to T_2 and ρ . This explains why T_1 remains the least accurately recovered parameter in the reduced

reconstruction experiments. The same analysis also shows coupling between T_2 and ρ , visible both in the normalized Hessian correlation matrix and in the tilted contour plots in the (T_2, ρ) -plane. The weak T_1 sensitivity has a physical interpretation in the simplified measurement model. The measured signal is proportional to $\rho(m_x + im_y)$, so it directly observes only the transverse components of the magnetization. The parameter T_2 directly affects the transverse decay, and ρ directly rescales the measured signal amplitude. In contrast, T_1 mainly controls the longitudinal component m_z , which is not measured directly. Information about T_1 therefore enters the measured signal only indirectly through the coupled Bloch dynamics. For the chosen initial condition and observation window, this indirect influence is weak. The remaining reconstruction error is therefore not simply a failure of the numerical method, but reflects the local identifiability structure of the experiment. The noise experiments in Chapter ?? support the same interpretation. Adding relative complex-valued noise increases the variability of the reconstructed parameters, but the dominant variability appears in the T_1 component. The mean reconstruction error remains close to the zero-noise baseline over the tested noise levels, while the standard deviation increases with the noise level. Thus, in this experiment, the main limitation is not the added noise alone. Rather, noise is amplified most strongly in the same weak direction already identified by the Hessian analysis.

Limitations Several limitations should be kept in mind when interpreting these results. First, the model is a simplified single-voxel model. Spatial encoding, multiple receiver coils, realistic pulse sequences, and spatially varying parameter fields are not included. This makes the problem useful for isolating the mathematical effects of discretization, penalization, and sensitivity, but it is not yet a full quantitative MRI reconstruction model. Second, the effective magnetic field is taken to be constant in the numerical experiments. In realistic MRI acquisitions, the applied fields are time-dependent and include RF pulses and gradients. Including such effects would change the Bloch dynamics and could also change the sensitivity of the data to the parameters. Also, the time discretization is based on Forward Euler. This choice is useful because its stability and discretization error are easy to interpret, but it is not the most accurate or stable method for oscillatory Bloch dynamics. The large errors for time steps above the stability threshold are therefore partly a consequence of using an explicit first-order method. This also brings another point, the penalty formulation is solved numerically with a finite number of iterations. The formal penalty principle describes what should happen as $\lambda \rightarrow \infty$ for sufficiently accurate minimization, but the computations are also affected by finite λ , finite iteration count, nonconvexity, and the conditioning of the optimization problem. Therefore, discrepancies between the large- λ penalty solution and the reduced solution should not be interpreted as a failure of the penalty method alone. Finally, the sensitivity and Hessian analysis is local. It describes the geometry of the objective near the true parameter vector for the chosen initial condition and observation window. It does not rule out different behaviour for other initial magnetizations, longer acquisition windows, different fields, or multi-experiment designs.

Future directions Several extensions would be natural, as indicated above. A first direction is to replace Forward Euler by a more stable or higher-order time integrator. This would reduce the influence of time discretization and make it possible to separate model sensitivity from numerical instability more cleanly. Then, using analytic sensitivities or adjoint-based derivatives instead of finite differences. This would make the reconstruction algorithms more efficient and would also allow a more systematic comparison between reduced, all-at-once, and penalized optimization methods. Third note is to improve the experimental design. Since the main difficulty in the present setup is weak sensitivity to T_1 , one could choose different initial magnetizations, longer observation windows, or multiple excitation experiments to increase the information content of the measured signal. The Hessian diagnostics used in this thesis could then be used to compare which designs improve identifiability. Overall, the results of this thesis show that parameter reconstruction in the Bloch inverse problem is governed by more than numerical accuracy alone. A small time step and a small ODE residual are important, but they do not guarantee accurate parameter recovery if the measured signal is weakly sensitive to some parameter directions. For the single-voxel experiment studied here, the dominant difficulty is the weak local identifiability of T_1 . Understanding and improving this sensitivity structure is therefore essential for reliable quantitative MRI reconstruction.

7 Appendix

Algorithm 1 Reduced Gauss–Newton for the Bloch ODE

1: **Input:** initial guess $\mathbf{p}^{(0)}$, step size h , data $\{d_n\}$, tolerance ε , maximum iterations $K_{\max}$.

2: **for** $k = 0, 1, \dots, K_{\max} - 1$ **do**

3: **Forward solve:** Compute $\{M_n\}_{n=0}^N$ from

$$M_{n+1} = M_n + h(A(\mathbf{p}^{(k)})M_n + q(\mathbf{p}^{(k)})).$$

4: Compute residuals $r_n = CM_n - d_n$ and objective

$$J(p^{(k)}) = \sum_{n=0}^{N-1} \|r_n\|_2^2.$$

5: **Adjoint solve:** Set $\lambda_{N-1} = 2r_{N-1}$ and for $n = N-2, \dots, 0$ compute

$$\lambda_n = (I_3 + hA(p^{(k)}))^\top \lambda_{n+1} + 2r_n.$$

6: **Gradient evaluation:** For $j = 1, 2, 3$,

$$g_j^{(k)} = h \sum_{n=0}^{N-1} \lambda_n^\top \left(\frac{\partial A}{\partial p_j} M_n + \frac{\partial q}{\partial p_j} \right).$$

7: **Stopping test:** If $\|g^{(k)}\|_2 < \varepsilon$, terminate.

8: **Gauss–Newton matrix (finite-difference approximation):** Approximate $H^{(k)} \approx \nabla^2 J(p^{(k)})$ by symmetric finite differences of g .

9: **Parameter update:** Solve

$$H^{(k)} \Delta p^{(k)} = -g^{(k)},$$

and set

$$p^{(k+1)} = p^{(k)} + \alpha \Delta p^{(k)}.$$

10: **end for**

11: **Output:** estimated parameters $p^{(k+1)}$.

Algorithm 2 Penalized Gauss–Newton for the Bloch ODE

1: **Input:** initial parameter guess $\mathbf{p}^{(0)}$, initial trajectory $\{M_n^{(0)}\}_{n=0}^N$, step size h , data $\{d_n\}$, penalty parameter λ , tolerance ε , maximum iterations $K_{\max}$.

2: **for** $k = 0, 1, \dots, K_{\max} - 1$ **do**

3: **Residual evaluation:** Compute the data residuals

$$r_n^{(k)} = CM_n^{(k)} - d_n, \quad n = 0, \dots, N,$$

and the discrete ODE residuals

$$R_n^{(k)} = \frac{M_{n+1}^{(k)} - M_n^{(k)}}{h} - (A(\mathbf{p}^{(k)})M_n^{(k)} + q(\mathbf{p}^{(k)})), \quad n = 0, \dots, N-1.$$

4: Compute the penalized objective

$$J_{h,\lambda}(M^{(k)}, \mathbf{p}^{(k)}) = \frac{h}{2} \sum_{n=0}^N \|r_n^{(k)}\|_2^2 + \frac{\lambda h}{2} \sum_{n=0}^{N-1} \|R_n^{(k)}\|_2^2.$$

5: **Gradient evaluation:** Compute the gradients with respect to the state trajectory and the parameters,

$$g_M^{(k)} = \nabla_M J_{h,\lambda}(M^{(k)}, \mathbf{p}^{(k)}), \quad g_p^{(k)} = \nabla_p J_{h,\lambda}(M^{(k)}, \mathbf{p}^{(k)}).$$

Combine them into the full gradient

$$g^{(k)} = \begin{pmatrix} g_M^{(k)} \\ g_p^{(k)} \end{pmatrix}.$$

6: **Stopping test:** If $\|g^{(k)}\|_2 < \varepsilon$, terminate.

7: **Gauss–Newton matrix (finite-difference approximation):** Approximate

$$H^{(k)} \approx \nabla^2 J_{h,\lambda}(M^{(k)}, \mathbf{p}^{(k)})$$

by symmetric finite differences of g with respect to both M and $\mathbf{p}$.

8: **Joint state-parameter update:** Solve

$$H^{(k)} \begin{pmatrix} \Delta M^{(k)} \\ \Delta p^{(k)} \end{pmatrix} = - \begin{pmatrix} g_M^{(k)} \\ g_p^{(k)} \end{pmatrix}.$$

9: Update the state trajectory and parameter vector:

$$M^{(k+1)} = M^{(k)} + \alpha \Delta M^{(k)},$$

$$p^{(k+1)} = p^{(k)} + \alpha \Delta p^{(k)}.$$

10: **end for**

11: **Output:** estimated trajectory $M^{(k+1)}$ and estimated parameters $p^{(k+1)}$.

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